Prequential Streaming Evaluation of Stateful Time-Series Forecasters with Online Conformal Calibration: A Diagnostic Atlas and the `preqts` Harness

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Not peer reviewed. Source code, data and computational records:
`github.com/fsantibanezleal/CAOS_RES_ChronoScope`

***Abstract*. Forecasting may be deployed as a stream: observations arrive one by one, the model updates, and its prediction intervals are expected to remain useful as the series changes. We provide a prequential streaming evaluation harness, released as the open-source package `preqts`: it implements Dawid's predict-then-observe-then-update protocol behind a `StatefulForecaster` interface with an explicit covariate-arrival policy, and records four trajectories per method: rolling scaled error, rolling empirical coverage, rolling weighted quantile loss, and cumulative measured compute time. The package supports stateful and replay-based forecasters; the experiment reported here uses replay-based Theta wrappers and does not compare constant-cost native state updates. The harness is embedded in an atlas that characterizes each series with ten diagnostic families (stationarity, autocorrelation, seasonality, filters, change points, volatility, distribution and complexity, fractal structure, nonlinear dynamics, and cross-series causality) and then forecasts it with a method catalogue spanning classical, statistical, gradient-boosted, deep, and foundation-model methods. Using the harness we test a concrete claim: that online conformal calibration keeps streaming intervals at nominal under distribution shift. On fifteen series that span distinct regimes and include white-noise and random-walk negative controls, wrapping a Theta point forecaster in Adaptive Conformal Inference or Conformal-PID control reduces the mean absolute coverage error at the nominal 80% level from 0.0416 (raw Theta) and 0.0599 (seasonal-naive) to 0.0057 and 0.0060, a seven-to-tenfold reduction, and shrinks the across-case span of the final coverage from 0.19–0.40 to 0.04, without changing the point forecast. Calibrated coverage is also near target on the two generated negative controls, which provide limited checks against simple predictable-versus-unpredictable confounds but cannot rule out all leakage. The contribution is the reusable prequential harness and an artifact-backed comparison under the protocol defined below.**

***Index Terms*. time-series forecasting, prequential evaluation, streaming, conformal prediction, adaptive conformal inference, calibration, coverage, stateful forecasters, foundation models, diagnostics, benchmark, reproducibility.**

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The harness is released as the open-source package `preqts` (https://pypi.org/project/preqts/); the atlas (code, per-case artifacts, workbench) is at https://github.com/fsantibanezleal/CAOS_RES_ChronoScope (MIT), and an interactive workbench is available at https://chronoscope.fasl-work.com.

I. Introduction

FORECASTING deployed as a stream has a sequential information constraint: at each cutoff, a forecast must be made from information available then, scored after the target arrives, and followed by an update. Rolling-origin evaluations are useful for other questions; their treatment of model state depends on the evaluated method and adapter. We do not claim that existing benchmark harnesses universally omit stateful or prequential evaluation. We instead describe and release `preqts`, which makes the predict-score-ingest order and covariate arrival policy explicit, and use it to evaluate an artifact-backed interval-calibration study.

The harness implements the prequential principle, Dawid's "predictive sequential" protocol [1]: at each step the model predicts, the truth is revealed and scored, and only then does the model update. Metrics can therefore be inspected along the stream rather than only as one aggregate. The harness is embedded in an atlas (Section II) that pairs forecasts with a diagnostic layer. We then use the harness to measure (Section V): does online conformal calibration keep streaming intervals near the nominal level on this selected set of changing and control series? Our contributions are: (i) the reusable prequential streaming harness `preqts`, with a covariate-arrival policy; and (ii) an artifact-backed comparison showing lower descriptive coverage error for online calibration than for raw intervals in these 15 cases, while retaining the same Theta median forecast.

II. The atlas: understand, then forecast

ChronoScope pairs a diagnostic layer with a forecast ladder over fifteen cases, each computed deterministically (an artifact is a pure function of case and seed). **Ten diagnostic families** characterize a series before it is forecast: stationarity and

unit roots (ADF, KPSS, Phillips–Perron, DF-GLS, Zivot–Andrews); autocorrelation (ACF/PACF, Ljung–Box); seasonality (periodogram, STL/MSTL, seasonal strength); filters and adaptive decompositions (Hodrick–Prescott, Baxter–King, EMD/CEEMDAN, wavelet scalogram); change points and regimes (PELT, OLS-CUSUM, Hamilton Markov-switching); volatility and transforms (ARCH-LM, GARCH, Box–Cox); distribution, normality and complexity (moments, Jarque–Bera, sample/spectral entropy, the BDS test, and the catch22 feature set [9]); fractal and multifractal structure (Hurst via R/S and DFA, MF-DFA, ARFIMA long memory); nonlinear dynamics and chaos (Takens embedding, largest Lyapunov exponent, recurrence quantification, the 0–1 test, with a surrogate-data test for nonlinearity); and cross-series causality (cross-correlation lead/lag, Granger, Engle–Granger and Johansen cointegration). Each is a verified implementation wrapped in a NaN-safe API and precomputed per case.

A nineteen-method forecast ladder is then scored on the same cases: classical numpy live methods (seasonal-naive, SES, Holt, Holt–Winters, Theta [8]), statistical methods (AutoARIMA, AutoETS, AutoTheta), a gradient-boosted method (LightGBM), deep methods, and stateful zero-shot foundation models (Chronos [10], TimesFM [11]). The cases span distinct regimes by construction, level shift, deterministic chaos, GARCH heteroskedasticity, intermittent demand, long memory, coupled daily/weekly seasonality, exogenous promotions, plus real CC-BY series (UCI Electricity, Beijing PM2.5, M4 excerpts [6]) and two negative controls, white noise and a near-random walk. There is no universal winner: the diagnosis explains the leaderboard, which is exactly the setting in which leakage-free scoring matters most.

III. The prequential streaming protocol

Let the stored target series be $y_0, \dots, y_{T-1}$, let m denote its case-specific seasonal period, and let h denote the forecast horizon. In the experiment reported here $h = 1$ and the cutoff step is one observation. At cutoff t, the forecaster receives only the initial context $y_{0:t-1}$ and observations subsequently revealed at earlier cutoffs. It returns quantiles at $q \in \{0.1, 0.5, 0.9\}$. The prediction is recorded and scored against the next observation before that observation is passed to the forecaster:

$$(\hat{Q}_{t,0.1}, \hat{Q}_{t,0.5}, \hat{Q}_{t,0.9}) = f_{\text{predict}}(s_t), \qquad s_{t+1} = f_{\text{ingest}}(s_t, y_t). \tag{1}$$

The initial context length is $w = \min(\max(2m, 20), \max(1, \lfloor T/3 \rfloor))$. The harness initializes each forecaster once with $y_{0:w-1}$ and evaluates cutoffs $t = w, \dots, T-h$; after scoring at t, it ingests y_t and advances by one. Thus the target being scored is not available to either the point forecaster or the interval controller when the forecast is made. For a covariate channel, the stream contract distinguishes past-only values (optionally delayed by their declared lag) from known-future values. The conformal experiment has no covariates, so this policy does not affect its results.

The evaluated point method is the repository's NumPy Theta implementation, not a streaming-native Theta state update. At every cutoff, a `ReplayAdapter` calls the batch forecaster on the full history then available. For a context $x_0, \dots, x_{n-1}$, the implementation estimates an OLS slope b against integer time, selects the SES coefficient a from $\{0.05, 0.10, \dots, 0.95\}$ by minimum in-sample one-step squared error, and updates the level $\ell_i = ax_i + (1-a)\ell_{i-1}$ from $\ell_0 = x_0$. Its one-step fitted values are the prior levels; $\hat{\sigma}$ is the population standard deviation of the resulting one-step residuals. For lead $j = 1, \dots, h$, the point path is $\hat{y}_{t+j} = \ell_{n-1} + 0.5bj$. The base quantiles are $\hat{Q}_{t,j,q} = \hat{y}_{t+j} + \Phi^{-1}(q)\hat{\sigma}\sqrt{j}$, using the implementation's Acklam approximation to the normal quantile. This Gaussian widening is an implementation choice, not a distribution-free interval guarantee. In the reported one-step experiment, this reduces to the current Theta point forecast plus or minus $\Phi^{-1}(0.9)\hat{\sigma}$.

At each forecast the harness records one-step MASE, empirical coverage of the inclusive interval $[\hat{Q}_{0.1}, \hat{Q}_{0.9}]$, WQL at the three requested quantiles, and prediction and ingestion latency. MASE divides absolute error by a single scale fixed from the warmup context, $S_w = (T_w - m)^{-1} \sum_{i=m}^{w-1} |y_i - y_{i-m}|$ (with the package's documented fallback for a short or constant context). WQL follows the implementation's GluonTS-style normalization: $2\sum_q \rho_q(y - \hat{Q}_q) / \sum_i |y_i|$, where $\rho_q(u) = u(q - \mathbf{1}\{u < 0\})$ is pinball loss. Each displayed rolling trajectory is the trailing arithmetic mean over at most 25 cutoff values, including the current cutoff. Cumulative cost is the running sum of measured prediction and ingestion latencies; it is a machine- and run-dependent timing observation, not an asymptotic complexity proof. The package also computes other metrics, but they are not used for the headline plots in this work. The three-quantile WQL summary is not treated as a CRPS estimate.

The experiment uses the package API `run_prequential` from `preqts==0.3.0`; the series, case-specific seasonal periods and generated artifacts are those committed under `data/derived/`. Each method starts from the same case prefix and receives the same target sequence. The four methods are seasonal-naive, uncalibrated Theta, Theta with ACI, and Theta with Conformal-PID. Only the two wrappers change the interval; they retain the same Theta median forecast. The adapters recompute each batch forecast from its accumulated history, so this experiment compares their interval behavior under a common replay protocol and does not measure the benefit of persistent internal model state.

IV. Interval calibration and experimental design

The wrappers recalibrate the outer interval of the same Theta forecast. For a one-step base forecast, write its median as m_t and its half-width as $r_t = \max\{(\hat{Q}_{t,0.9} - \hat{Q}_{t,0.1})/2, 10^{-8}\}$. Once y_t arrives, the nonconformity score is the scale-normalized absolute error $z_t = |y_t - m_t|/r_t$. This is not raw-residual split conformal: it is a sequential score buffer and a multiplicative correction to the base interval. Each horizon has a separate state in the package; the experiment uses only horizon one. For either method, the calibrated outer endpoints are $m_t \pm \hat{q}_t r_t$, where $\hat{q}_t$ is the empirical quantile of

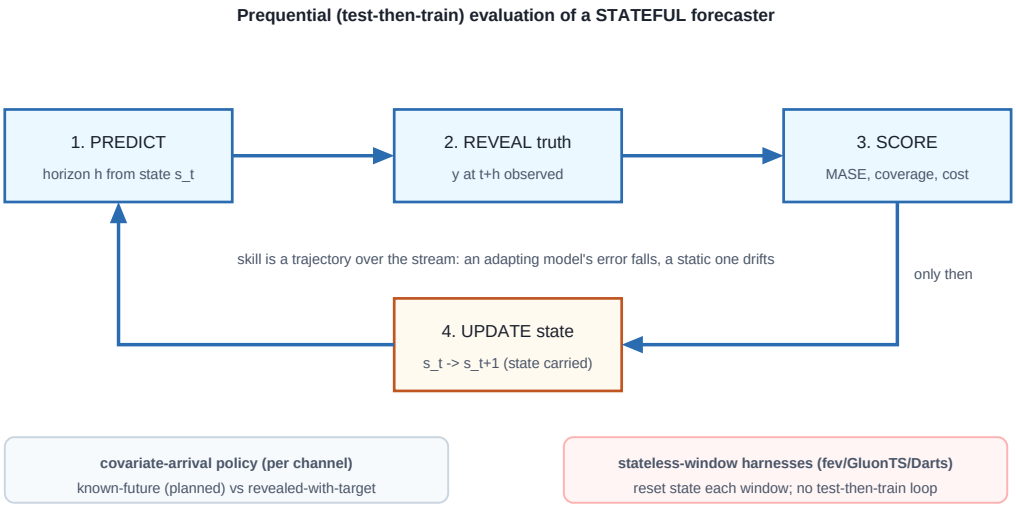


Fig. 1. The prequential test-then-train loop for a stateful forecaster. At each step the model predicts from its current state, the truth is revealed and scored, and only then is the state updated; a covariate-arrival policy declares, per channel, whether a value is known at prediction time or revealed with the target. Stateless-window harnesses reset state each window and run no test-then-train loop, so they cannot see a calibrator recover coverage after a shift.

the previous scores in that horizon's buffer at level $1-\alpha_t$. The buffer is a FIFO deque capped at 200 scores and the quantile is NumPy's default linearly interpolated sample quantile. Before any score is available, $\hat{q}_t = 1$. The current prediction is therefore formed from past state only; its score enters the buffer after that prediction has been scored.

Adaptive Conformal Inference (ACI) [2] begins with target miscoverage $\alpha_0 = 1-c = 0.2$, where $c = 0.8$. After the current miss indicator $e_t = \mathbf{1}\{|y_t - m_t| > \hat{q}_t r_t\}$ is evaluated using the multiplier that generated the forecast, it updates

$$\alpha_{t+1} = \mathrm{clip}\big(\alpha_t + \gamma\big((1-c) - e_t\big), 10^{-4}, 1-10^{-4}\big)\,, \quad (2)$$

with $\gamma = 0.02$. A miss reduces α_t, raising the next quantile level and generally expanding the interval; a covered observation increases α_t and generally contracts it. Clipping bounds the effective level. These implementation details do not establish the unqualified convergence claim stated in earlier drafts: the empirical rolling-quantile buffer, level clipping, and this particular base interval define the evaluated procedure, and the results below are finite-stream measurements rather than a new coverage guarantee.

Conformal-PID [3] uses the same score buffer and quantile rule. With target miscoverage $\alpha^* = 0.2$, its error is $u_t = \alpha^* - e_t$, its integral is $I_t = I_{t-1} + u_t$ (initialized at zero), and its derivative is $d_t = u_t - u_{t-1}$ (initialized to zero on the first update). The controller computes

$$v_t = k_p u_t + k_i I_t + k_d d_t, \quad (3)$$

$$\alpha_{t+1} = \mathrm{clip}(\alpha^* + v_t, 10^{-4}, 1-10^{-4}), \quad (4)$$

with $k_p = 0.05$, $k_i = 0.02$, and $k_d = 0.02$. The cumulative integral itself is not separately clipped in the implementation. These gains are hard-coded in the bake; the repository does not record a parameter-selection or sensitivity-analysis procedure. The wrapper changes only the outer quantiles, leaving the median column and hence the point forecast unchanged.

The comparison is a fixed artifact-based study of 15 repository cases: 11 generated regimes and four redistributed real-data cases, with two generated cases designated as negative controls (white noise and near-random walk). The checked-in artifacts identify the cases and preserve their streaming metric trajectories; underlying series values are not included for every source. The streaming stage sets $c = 0.8$, $h = 1$, levels $(0.1, 0.5, 0.9)$, step one, and the warmup formula in Section III. The main figures give equal weight to each case. For case i and method a, let $C_{i,a,k}$ be the stored trailing-25 coverage at cutoff k. The plotted summary is $\frac{1}{15}\sum_i |K_i^{-1}\sum_{k\in K_i} C_{i,a,k} - 0.8|$; the whisker reaches the largest case-level absolute deviation. The per-case figure instead plots each artifact's final full-run coverage, with its across-case range. From the committed trajectories, the rolling-trajectory summary is 0.0599 for seasonal-naive, 0.0416 for raw Theta, 0.0057 for Theta+ACI and 0.0060 for Theta+PID; the largest case-level deviations are 0.3071, 0.1124, 0.0225 and 0.0224, respectively. The final-coverage spans are 0.3964, 0.1912, 0.0444 and 0.0444, respectively. These are descriptive summaries of this selected set, not inferential estimates for a population of forecasting tasks.

The calibration result is limited in several ways. First, there are only 15 deliberately varied cases, and the four real-data examples do not establish broad real-world generalization. Second, the experiment evaluates a one-step horizon, one nominal coverage, one base forecaster and one fixed parameter setting per controller; no sensitivity or repeated-seed study is present. Third, the base interval is a Gaussian approximation using in-sample residual dispersion, and Theta is re-fit by full-history replay at every cutoff. Fourth, the committed aggregate trajectories do not by themselves preserve per-time forecasts, scores, controller states, or timing environment metadata, so they permit the reported aggregate calculations but not an independent replay of every online update from the artifact

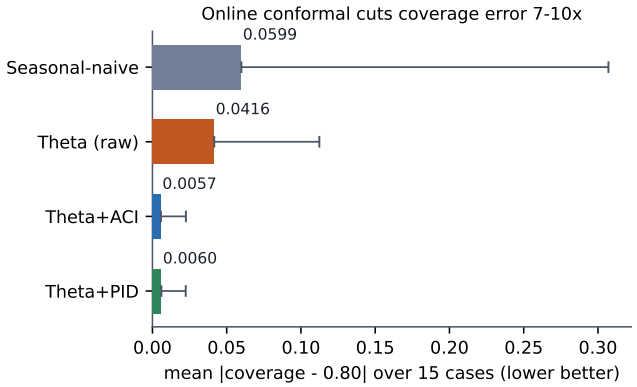


Fig. 2. Mean absolute coverage error at nominal 80% over the fifteen cases (bar) with the worst-case per-case error (whisker). Online conformal cuts the error seven-to-tenfold. Data: the per-case `streaming.json` artifacts.

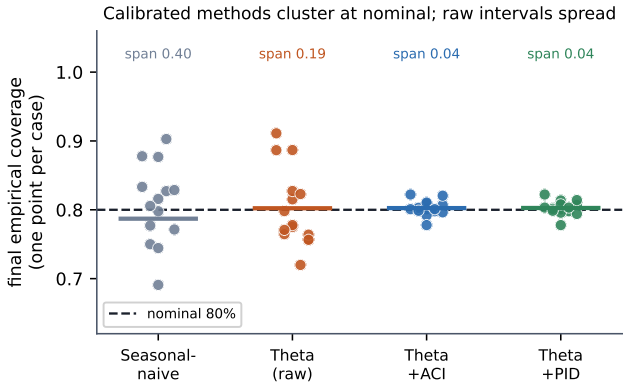


Fig. 3. Final empirical coverage, one point per case. The calibrated methods cluster tightly at the nominal level (across-case span 0.04) while the raw and naive intervals spread widely (span 0.19 and 0.40). Data: `streaming.json`.

alone. Finally, finite-run calibration and the two controls do not prove distribution-free coverage, rule out every form of leakage, or establish a general advantage over other forecasting methods. The exact rationale for the hard-coded ACI/PID gains and the complete source-data revision identifiers are not recorded in the streaming artifacts; those provenance gaps remain open.

V. Results

The headline is Fig. 2. Averaged over the fifteen cases, the raw Theta intervals miss the nominal 80% target by a mean absolute coverage error of 0.0416 and the seasonal-naive floor by 0.0599, with worst-case per-case errors of 0.1124 and 0.3071. Wrapping the *same* Theta point forecaster in ACI or Conformal-PID cuts the mean coverage error to 0.0057 and 0.0060, a sevenfold reduction over raw Theta and tenfold over seasonal-naive. The across-case view (Fig. 3) shows the same effect as a collapse of dispersion: the final coverage of the calibrated methods clusters within a 0.04 span at nominal, while raw Theta spreads over 0.19 and seasonal-naive over 0.40. The point forecast is unchanged (the MASE of Theta, Theta+ACI and Theta+PID is identical per case); only the interval self-corrects. The effect is regime-robust: where raw intervals drift most, the calibrators pull the time-averaged coverage back (intermittent demand from 0.91 raw to 0.81 with ACI; chaotic Mackey–Glass from 0.77 raw to 0.80; real UCI electricity from 0.76 raw to 0.80). On the white-noise and near-random-walk controls, observed coverage summaries are also near nominal after calibration. These two examples are limited checks against simple confounds; they do not establish that the harness is free of leakage.

VI. Discussion and Scope

Online calibration is designed to adapt interval coverage, and the measurements here concern one specific replay-based implementation and selected set of cases. The prequential protocol scores each forecast before revealing its target, while the full-history replay makes the base Theta forecast and its timing depend on the accumulated context. The identical median forecasts across raw and calibrated variants isolate interval adjustment in this experiment; the negative controls provide only two limited checks and do not rule out leakage. The atlas combines generated regimes with four real-data cases, so its real-data coverage evidence remains narrow. Results are shown at horizon one and nominal 80%; the package interface accepts other horizons and coverage levels, but those settings are not evaluated here. This work does not propose a new calibrator or establish a general advantage over alternative forecasting methods.

VII. Conclusion

We describe `preqts`, a prequential evaluation harness with an explicit covariate-arrival policy, and report an artifact-backed interval-calibration comparison on 15 selected cases. Under the stated one-step protocol, the mean absolute deviation of each case's rolling coverage summary from the nominal level is lower for ACI and Conformal-PID than for the uncalibrated Theta and seasonal-naive intervals. The median forecast is unchanged by the wrappers. These finite-case results do not establish distribution-free coverage or generalization to other horizons, base forecasters, controller settings, or data populations.

Data and Code Availability

The harness is the open-source package `preqts` (https://pypi.org/project/preqts/). The atlas code, the per-case artifacts (trace, analysis, and streaming trajectories) and the workbench are at https://github.com/fsantibanezleal/CAOS_RES_ChronoScope under the MIT license; an interactive workbench is available at https://chronoscope.fasl-work.com. Real datasets (UCI Electricity, Beijing PM2.5, M4 excerpts) are distributed under CC-BY-4.0 with attribution; streaming artifacts carry aggregate trajectories only. The figures are regenerated deterministically from these artifacts by `manuscripts/prequential-streaming-conformal/figures/make_figs.py`.

APPENDIX A
THE ATLAS ROSTER

The ten diagnostic families and their principal methods are listed in Section II; each is a verified implementation with its own theory page (equations, DOIs, and a theme-aware figure) in the repository `docs/analysis/`. The nineteen-method forecast ladder spans classical (seasonal-naive, SES, Holt, Holt–Winters, Theta), statistical (AutoARIMA, AutoETS, AutoTheta), gradient-boosted (LightGBM), deep, and stateful foundation models (Chronos, TimesFM). The streaming roster reported here is the naive floor (seasonal-naive), a strong classical live method (Theta) with raw intervals, and that same Theta wrapped in ACI and in Conformal-PID; all four are evaluated by the prequential harness of Section III at nominal coverage 80%, horizon one, and a per-case warmup of twice the seasonal period floored at 20 steps (20, 24 or 48 across the cases), on all fifteen cases.

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